

Yuxuan Zhang

(65) 8941-6785
e1216649@u.nus.edu

yuxuanzhang271.github.io
github.com/yuxuanzhang271
linkedin.com/in/yuxuanzhang271

Skills

Coding: Python, C/C++, Bash (Shell), SQL, MATLAB, R, Verilog, Java, C#, JavaScript/TypeScript
Tools: Linux, Git, CMake, Docker, CUDA, ROS/ROS2, PyTorch, TensorFlow, OpenCV, PCL, Open3D, Isaac Gym/Sim, MuJoCo, Unity, LabVIEW, Altium Designer, AutoCAD, SolidWorks, Multisim, Keil μ Vision, KiCad
Languages: Chinese (native), English (fluent)

Education

Master in Artificial Intelligence Systems Aug 2025 – Present
National University of Singapore Singapore

- GPA: *TBD*
- Coursework: Robotic Systems, Natural Language Processing, Pattern Recognition, Machine Reasoning

Graduate Diploma in System Analysis Jul 2023 – Jan 2025
National University of Singapore Singapore

- GPA: 4.11/5.00
- Coursework: Machine Learning, Full-Stack Development, System Design

Bachelor in Intelligent Test & Control Engineering Sep 2020 – Jun 2024
Harbin Institute of Technology China

- GPA: 4.02/5.00
- Coursework: AI Fundamentals, Pattern Recognition & Machine Learning, Measurement & Control Systems, Embedded Systems, Distributed Systems, Digital Signal Processing

Publications

1. Chongkai Gao, Zixuan Liu, Zhenghao Chi, Junshan Huang, Xin Fei, Yiwen Hou, **Yuxuan Zhang**, Yudi Lin, Zhirui Fang, Zeyu Jiang, Lin Shao. “VLA-OS: Structuring and Dissecting Planning Representations and Paradigms in Vision-Language-Action Models.” *NeurIPS 2025* (accepted). [arXiv:2506.17561](https://arxiv.org/abs/2506.17561).
 - Proposed VLA-OS, a modular VLA framework that unifies and compares Action-Only, Integrated, and Hierarchical planning for robotic manipulation.
 - Built multi-representation planning heads and validated on LIBERO/COLOSSEUM/DexArt, showing hierarchical visually grounded planning improves generalization and long-horizon performance.

Preprints

1. Chongkai Gao, Jieao Shi, Xuchuan Huang, Mu Zhaoyu, **Yuxuan Zhang**, et al. “ManiLadder: Benchmarking Manipulation Intelligence Frontier via a Categorized and Multi-Level Task Ladder.” *CoRL 2026*.
 - Proposed ManiLadder, a large-scale manipulation benchmark with 112 tasks across hierarchical difficulty levels.
 - Built a GPU-accelerated simulator and standardized evaluation pipeline for reproducible benchmarking.
2. **Yuxuan Zhang**, et al. “CCTA: Safety- and Capability-Constrained Tool Arbitration for Long-Horizon Embodied Decision Making.” *in preparation (target: IEEE)*.
 - Proposed a safety-constrained tool arbitration framework for embodied agents, integrating capability modeling and decision-making.
 - Formulated tool selection as a constrained optimization problem with control barrier function (CBF)-based safety guarantees.
 - Developed compute-aware scheduling to reduce unnecessary invocation of high-cost models.

3. Haziq Razali, **Yuxuan Zhang**, Qianli Xu, Yiannis Demiris. “Learning to Synthesize Novel Human-Object Interaction in Collaborative Task-Based Settings.” *AAAI 2026 Workshop HCM Submission*.
 - Built a multi-view RGB-D pipeline (4D-Humans, HaMeR, SAM-6D) for synchronized 3D body–hand–object reconstruction.
 - Proposed LLM-guided task decomposition for long-horizon human-object interaction synthesis.

Selected Experience

Research Intern (Hardware / Software / System Design) *Mar 2026 – Aug 2026*
AMD Singapore Singapore

- Built a closed-loop LeRobot SO-101 pipeline from Web UI-based data collection to policy fine-tuning (ACT, π_0) and real-world inference.
- Optimized inference performance on AMD Radeon 8050S iGPU and enabled a Xilinx on-chip low-latency control system for real-time deployment.

AI Engineer Intern *Oct 2025 – Dec 2025*
Fatfish AI Singapore

- Integrated a humanoid system with MiniPC, RealSense, and Jetson Orin Nano via a unified ROS/ROS2 pipeline.
- Implemented perception–language–control integration with SLAM, object tracking, and RL/IK/VLA policies.
- Validated sim-to-real transfer on physical hardware.

Research Intern (Assistive Robotics) *Jun 2025 – Sep 2025*
AdaComp Lab, National University of Singapore Singapore

- Built a quadruped assistive system with LiDAR, panoramic vision, and onboard compute.
- Developed LiDAR-based navigation and panoramic object detection for user guidance via multimodal feedback.

Research Intern (Human-Behaviour Understanding) *Mar 2025 – Dec 2025*
*I2R, A*STAR* Singapore

- Proposed a data generation framework combining multi-view RGB-D capture with LLM-based task reasoning.
- Built a 3D reconstruction pipeline and enabled long-horizon HOI synthesis via task decomposition.

R&D Intern (Robotics Vision System) *Jul 2021 – Aug 2021*
Changjiang Intelligent Control Co., Ltd. China

- Developed a stereo vision pipeline for 3D reconstruction, pose estimation, and laser-based surface processing.

Tech Art Intern *Jan 2021 – Feb 2021*
Boke City Co., Ltd. (Wuhan) China

- Designed Unity-based VFX systems and implemented interaction logic in C#.

Selected Projects

Automated Calibration System for A/D Modules *Oct 2023 – May 2024*
Undergraduate Thesis, Harbin Institute of Technology China

- Built a 16-channel PCB with RS-232 firmware and integrated PXIe + LabVIEW for automated calibration.

Automated Drone Docking & Charging Platform *Oct 2021 – Sep 2022*
Innovation Project, Harbin Institute of Technology China

- Developed an autonomous docking system using GPS + AprilTag localization with closed-loop landing control.

3D Port Reconstruction *Oct 2020 – Sep 2021*
Project, Harbin Institute of Technology China

- Reconstructed large-scale 3D environments in Unity with VR interaction and dynamic simulation.